Lesson Assessment

First Measuring Conceptual Understanding

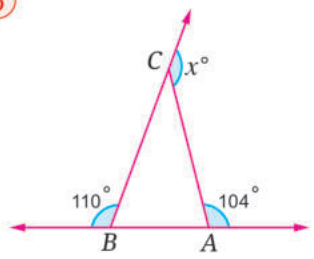
▸ **Choose the correct answer from the given ones:**

1. If the sum of the measures of two angles in a triangle is 130°, what is the measure of the third angle?
   (a) 20° (b) 30° (c) 50° (d) 60°
2. If the measures of two angles in a triangle are 30° and 70°, which of the following cannot be the measure of an exterior angle of the triangle?
   (a) 150° (b) 130° (c) 110° (d) 100°
3. Which of the following sets of numbers cannot be used as side lengths of a triangle?
   (a) 4 cm, 7 cm, 7 cm (b) 3 cm, 4 cm, 7 cm
   (c) 7 cm, 7 cm, 7 cm (d) 9 cm, 7 cm, 5 cm
4. The lengths of two sides of an isosceles triangle are 3 cm and 7 cm. What is the length of the third side?
   (a) 3 cm (b) 4 cm (c) 5 cm (d) 7 cm
5. If ABC is a scalene triangle in which the length of $\overline{AC}$ is 3 cm, and the length of $\overline{BC}$ is 5 cm. How many integers could be the length of $\overline{AB}$?
   (a) 2 (b) 3 (c) 4 (d) 5

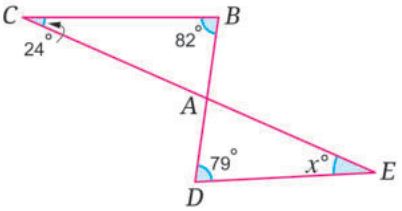
Second Applying Scientific Concepts

▸ **Find with proof the value of x in each of the following figures:**

6.

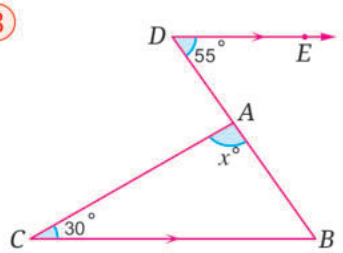


7.



$\overline{BD} \cap \overline{CE} = \{A\}$

8.

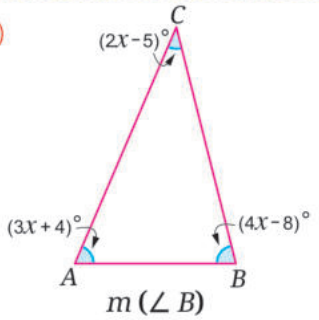


9. In the given picture, a wire of length 48 cm. The ends of this wire are bent at two points to form a triangle. Which of the following lengths could be bent to form a triangle with the remaining part?
   (a) 12 cm and 16 cm (b) 12 cm and 12 cm



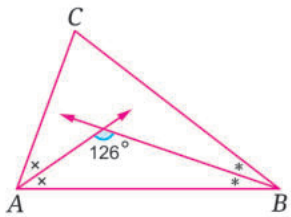
▸ **Find with proof the value of the required angles in each of the following figures:**

10.



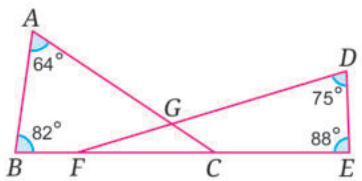
$m(\angle B)$

11.



$m(\angle C)$

12.



$m(\angle FGC)$